Introduction

A *combinatorial design*, or simply a *design*, is an arrangement of the objects of a set into subsets satisfying certain prescribed properties.

1. Perfect covers by dominoes of boards with forbidden positions where we arrange the allowed squares into pairs so that each pair can be covered by one domino.
2. Magic squares where we arrange the integers from 1 to n^2 in an n-by-n array so that certain sums are identical.
3. Latin squares where we arrange the integers 1 to n in an n-by-n array so that each integer occurs once in each row and once in each in each column.



Modular arithmetic I

- Let Z denote the set of integers

$$\{\ldots, -2, -1, 0, 1, 2, \ldots\},$$

and let $+$ and $\times$ denote ordinary addition and multiplication of integers.

- Let n be a positive integer with $n \geq 2$, and let

$$Z_n = \{0, 1, \ldots, n-1\}$$

be the set of nonnegative integers that are less than n. We can think of the integers in Z_n as the possible remainders when *any* integer is divided by n, *If m is an integer, then there exist unique integers q (the quotient) and r (the remainder) such that*

$$m = q \times n + r, \quad 0 \leq r \leq n-1.$$



Modular arithmetic II

- We define an addition, denoted $\oplus$, and a multiplication, denoted $\otimes$, on Z_n as follows,

  For any two integers a and b in Z_n, $a \oplus b$ is the (unique) remaind when the ordinary sum $a+b$ is divided by n, and $a \otimes b$ is the (unique) remainder when the ordinary product $a \times b$ is divided by n.

- Note that $\oplus$ and $\otimes$ depend on n, and we call them *addition mod n* and *multiplication mod n*, and with this addition and multiplication we get the *system of integer mod n*.



Modular arithmetic III

- We usually denote the arithmetic system of the integers mod n with the same symbol Z_n that we use for its set of elements.

Example

$n = 2,\ Z_2 = \{0, 1\}$

$\oplus$	0	1
0	0	1
1	1	0

$\otimes$	0	1
0	0	0
1	0	1

$n = 3,\ Z_2 = \{0, 1, 2\}$

$\oplus$	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

$\otimes$	0	1	2
0	0	0	0
1	0	1	2
2	0	2	1



Modular arithmetic IV

Example

Some instances of addition and multiplication in the system of integers modulo 6 are,

$$4 \oplus 5 = 3,$$
$$2 \oplus 3 = 5,$$
$$2 \otimes 2 = 4,$$
$$3 \otimes 5 = 3,$$
$$3 \otimes 2 = 0,$$
$$5 \otimes 5 = 1.$$

- The usual laws of commutativity, associativity, and distributivity are satisfied.
- An *additive inverse* of an integer a in Z_n is an integer b in Z_n such that $a \oplus b = 0$.



Modular arithmetic V

- If $a = 0$, then it's 0. if $a \neq 0$, then $n - a$ is between 1 and $n - 1$ and $n - a$ is an additive inverse of a. In all cases, the additive inverse is uniquely determined. Following usual conventions, it is denoted by $-a$.
- A *multiplicative inverse* of an integer a in Z_n is an integer b in Z_n such that $a \otimes b = 1$. We denote a multiplicative inverse of an integer a in Z_n by a^{-1}, *if there is one*.

Example

$n = 10$. the additive inverses are

$$-0 = 0 \quad -1 = 9 \quad -2 = 8 \quad -3 = 7 \quad -4 = 6$$
$$-5 = 5 \quad -9 = 1 \quad -8 = 2 \quad -7 = 3 \quad -6 = 4$$

and the situation with multiplicative inverses is the following

$$1^{-1} = 1 \quad 3^{-1} = 7 \quad 7^{-1} = 3 \quad 9^{-1} = 9.$$

None of 0, 2, 4, 5, 6, and 8 has a multiplicative inverse.

Modular arithmetic VI

- Note that if $-a = b$, then $-b = a$, put another way $-(-a) = a$.

Algorithm to compute the greatest common divisor (GCD) of two positive integers a and b

1. Set $A = a$ and $B = b$.
2. While $A \times B \neq 0$, do the following,
   - If $A \geq B$, then replace A by $A - B$.
   - Else, replace B by $B - A$.
3. Set GCD = B.

Lemma

The preceding algorithm terminates and computes the GCD of a and b correctly.



Modular arithmetic VII

- It is a consequence of this algorithm that the GCD, d, of a and b can be written as a linear combination of a and b with integral coefficients, there are integers x and y such that

$$d = a \times x + b \times y.$$

Example

Compute the GCD of 48 and 126.

A	48	48	48	18	18	6	6	0
B	126	78	30	30	12	12	6	6

A more efficient way $\rightarrow$

A	48	48	18	6	6
B	126	30	12	12	0

$6 = 8 \times 48 - 3 \times 126.$



Modular arithmetic VIII

Theorem

Let n be an integer with $n \geq 2$ and let a be a nonzero integer in $Z_n = \{0, 1, \ldots, n-1\}$. Then a has a multiplicative inverse in Z_n if and only if the greatest common divisor (GCD) of a and n is 1. If a has a multiplicative inverse, then it is unique.

Corollary

Let n be a prime number. Then each nonzero integer in Z_n has a multiplicative inverse.

- In addition, multiplicative inverses imply that the following properties hold in Z_n if n is a prime,
  1. (Cancelation rule 1) $a \otimes b = 0$ implies $a = 0$ or $b = 0$.
  2. (Cancelation rule 2) $a \otimes b = a \otimes c$, $a \neq 0$ implies $b = c$.
  3. (Solutions of linear equations) If $a \neq 0$, the equation $a \otimes x = b$ has the unique solution $x = a^{-1} \otimes b$.



Modular arithmetic IX

Example

Determine whether 11 has a multiplicative inverse in Z_{30}, and if so, calculate the multiplicative inverse.

A	B	
30	11	$30 = 2 \times 11 + 8$
8	11	$11 = 1 \times 8 + 3$
8	3	$8 = 2 \times 3 + 2$
2	3	$3 = 1 \times 2 + 1$
2	1	$2 = 2 \times 1 + 0$
0	1	$d = 1$



Modular arithmetic IX

Example

Determine whether 11 has a multiplicative inverse in Z_{30}, and if so, calculate the multiplicative inverse.

A	B		
30	11	$30 = 2 \times 11 + 8$	
8	11	$11 = 1 \times 8 + 3$	
8	3	$8 = 2 \times 3 + 2$	
2	3	$3 = 1 \times 2 + 1$	$1 = 3 - 1 \times 2$
2	1	$2 = 2 \times 1 + 0$	
0	1	$d = 1$	



Modular arithmetic IX

Example

Determine whether 11 has a multiplicative inverse in Z_{30}, and if so, calculate the multiplicative inverse.

A	B		
30	11	$30 = 2 \times 11 + 8$	
8	11	$11 = 1 \times 8 + 3$	
8	3	$8 = 2 \times 3 + 2$	$1 = 3 - 1 \times (8 - 2 \times 3) = 3 \times 3 - 1 \times 8$
2	3	$3 = 1 \times 2 + 1$	$1 = 3 - 1 \times 2$
2	1	$2 = 2 \times 1 + 0$	
0	1	$d = 1$	



Modular arithmetic IX

Example

Determine whether 11 has a multiplicative inverse in Z_{30}, and if so, calculate the multiplicative inverse.

A	B		
30	11	$30 = 2 \times 11 + 8$	
8	11	$11 = 1 \times 8 + 3$	$1 = 3 \times (11 - 1 \times 8) - 1 \times 8 = 3 \times 11 - 4 \times 8$
8	3	$8 = 2 \times 3 + 2$	$1 = 3 - 1 \times (8 - 2 \times 3) = 3 \times 3 - 1 \times 8$
2	3	$3 = 1 \times 2 + 1$	$1 = 3 - 1 \times 2$
2	1	$2 = 2 \times 1 + 0$	
0	1	$d = 1$	



Modular arithmetic IX

Example

Determine whether 11 has a multiplicative inverse in Z_{30}, and if so, calculate the multiplicative inverse.

A	B		
30	11	$30 = 2 \times 11 + 8$	$1 = 3 \times 11 - 4 \times (30 - 2 \times 11) = 11 \times 11 - 4 \times 30$
8	11	$11 = 1 \times 8 + 3$	$1 = 3 \times (11 - 1 \times 8) - 1 \times 8 = 3 \times 11 - 4 \times 8$
8	3	$8 = 2 \times 3 + 2$	$1 = 3 - 1 \times (8 - 2 \times 3) = 3 \times 3 - 1 \times 8$
2	3	$3 = 1 \times 2 + 1$	$1 = 3 - 1 \times 2$
2	1	$2 = 2 \times 1 + 0$	
0	1	$d = 1$	



Modular arithmetic IX

Example

Determine whether 11 has a multiplicative inverse in Z_{30}, and if so, calculate the multiplicative inverse.

A	B		$11^{-1} = 11$
30	11	$30 = 2 \times 11 + 8$	$1 = 3 \times 11 - 4 \times (30 - 2 \times 11) = 11 \times 11 - 4 \times 30$
8	11	$11 = 1 \times 8 + 3$	$1 = 3 \times (11 - 1 \times 8) - 1 \times 8 = 3 \times 11 - 4 \times 8$
8	3	$8 = 2 \times 3 + 2$	$1 = 3 - 1 \times (8 - 2 \times 3) = 3 \times 3 - 1 \times 8$
2	3	$3 = 1 \times 2 + 1$	$1 = 3 - 1 \times 2$
2	1	$2 = 2 \times 1 + 0$	
0	1	$d = 1$	



Modular arithmetic X

- The usual laws of arithmetic that we are accustomed to taking for granted in the arithmetic systems of real numbers or rational numbers also hold for Z_n, *provided n is a prime number*.
- There are other methods, however, to obtain finite arithmetical systems which satisfy the laws of arithmetic that we are accustomed to. The name given to such an arithmetical system is a *field*.
- Recall that the complex numbers are obtained from the real numbers by "adjoining" a root, usually denoted by i, of $i^2 + 1 = 0$. The system of complex numbers consists of all numbers of the form $a + bi$ where a and b are real numbers where the usual laws of arithmetic hold and $i^2 = -1$.
- At this point, we stop using the more cumbersome notation $\oplus$ and $\otimes$ for addition and multiplication mod n and use instead $+$ and $\times$, respectively.



Modular arithmetic XI

- This method can be used to construct fields with p^k elements for every prime p and integer $k \geq 2$ starting from the field Z_p.

Example

Construction of a field of 4 elements

We start with Z_2 and the polynomial $x^2 + x + 1$ with coefficients in Z_2. This polynomial has no root in Z_2. It can not be factored in any non-trivial way. We adjoin a root i of this polynomial to Z_2, getting $i^2 + i + 1 = 0$, or, equivalently,

$$i^2 = -i - 1 = i + 1.$$

The elements of the resulting field are the 4 elements

$$\{0, 1, i, 1 + i\},$$

with addition and multiplication tables given below,



Modular arithmetic XII

$+$	0	1	i	$1+i$
0	0	1	i	$1+i$
1	1	0	$1+i$	i
i	i	$1+i$	0	1
$1+i$	$1+i$	i	1	0

$\times$	0	1	i	$1+i$
0	0	0	0	0
1	0	1	i	$1+i$
i	0	i	$1+i$	1
$1+i$	0	$1+i$	1	i

Example

Construction of a field of $3^3 = 27$ elements

The polynomial $x^3 + 2x + 1$ with coefficients in Z_3 does not have a root in Z_3. Thus we adjoin a root i of this polynomial, getting $i^3 + 2i + 1 = 0$ or, equivalently, $i^3 = -1 - 2i = 2 + i$. The elements of the resulting field are 27-elements

$$\{a + bi + ci^2 : a,\ b \text{ and } c \text{ in } Z_3\}.$$



Modular arithmetic XIII

Some of the arithmetic are listed below,

$$
\begin{aligned}
(2+i+2i^2)+(1+i+i^2) &= 0+2i+0i^2 = 2i,\\
(1+i)\times(2+i^2) &= 2+i^2+2i+i^3 = 1+i^2,\\
(1+2i^2)\times(1+i+2i^2) &= 1+i+2i^2+2i^2+2i^3+4i^4\\
&= 1+i+i^2+1+2i+2i+i^2,\\
&= 2+2i+2i^2,\\
i^{-1} &= 1-2i^2,\\
(2+i+2i^2)^{-1} &= 1+i^2.
\end{aligned}
$$



Modular arithmetic XIV

- For each prime p and each integer $k \geq 2$ there exists a polynomial of degree k with coefficients in Z_p that does not have a nontrivial factorization. Thus, we can construct a field with p^k elements. Conversely, it can be proved that, if there is a field with a finite number m elements–that is, a finite system satisfying the usual rules of arithmetic, then $m = p^k$ for some positive integer k and some prime number p, and it can be obtained from Z_p (or is Z_p if $k = 1$). Thus, *only for a prime power number of elements do finite fields exist*.



Block designs I

Example

Suppose there are 7 varieties of a product to be tested for acceptability among consumers. The manufacturer plans to ask some random (or typical) consumers to compare the different varieties. The manufacturer decides to have each consumer do an incomplete test by comparing only some of the varieties. Thus, the manufacturer asks each person to compare a certain 3 of the varieties. In order to be able to draw meaningful conclusions based on statistical analysis of the results, the test is to have the property that each pair of the 7 varieties is compared by exactly one person. Can such a testing experiment be designed?

- We label the different varieties 0, 1, 2, 3, 4, 5 and 6. There are $\binom{7}{2} = 21$ pairs of the 7 varieties. Each tester gets 3 varieties and thus makes $\binom{3}{2} = 3$ comparisons. Since each pair is to be compared exactly one, the number of testers must be $21/3 = 7$.



Block designs II

- A collection of 7 blocks is

$$B_1 = \{0, 1, 3\}, B_2 = \{1, 2, 4\}, B_3 = \{2, 3, 5\}, B_4 = \{3, 4, 6\},$$
$$B_5 = \{0, 4, 5\}, B_6 = \{1, 5, 6\}, B_7 = \{0, 2, 6\}$$

It is presented in array

	0	1	2	3	4	5	6
B_1	1	1	0	1	0	0	0
B_2	0	1	1	0	1	0	0
B_3	0	0	1	1	0	1	0
B_4	0	0	0	1	1	0	1
B_5	1	0	0	0	1	1	0
B_6	0	1	0	0	0	1	1
B_7	1	0	1	0	0	0	1

varieties – v

k-element subsets – b blocks in total

occurrence of each pair of elements in blocks – index of the design – λ



Block designs III

- Let k, λ, and v be positive integers with $2 \leq k \leq v$. Let X be any set of v elements, called *varieties*, and let $\mathcal{B}$ be a collection $B_1, B_2, \ldots, B_b$ of k-element subsets of X called *blocks*. Then $\mathcal{B}$ is a *balanced block design* on X, provided that each pair of elements of X occurs together in exactly λ blocks. The number λ is called the *index of the design*.
- Let $\mathcal{B}$ be a balanced block design. If $k = v$, that is, the complete set of varieties occurs in each block, then the design $\mathcal{B}$ is called a *complete* block design. If $k < v$ and $\mathcal{B}$ is balanced, then we have a balanced *incomplete* block design, or *BIBD* for short.



Block designs IV

- Let $\mathcal{B}$ be a BIBD on X. We associate with $\mathcal{B}$ an *incidence matrix* or *incidence array* A. The array A has b rows, one corresponding to each of the blocks $B_1, B_2, \ldots, B_b$, and v columns, one corresponding to each of the varieties $x_1, x_2, \ldots, x_v$ in X. The array a_{ij} at the intersection of row i and column j is 0 or 1:

$$a_{ij} = 1 \quad \text{if } x_j \text{ is in } B_i, \qquad a_{ij} = 0 \quad \text{if } x_j \text{ is not in } B_i.$$

Lemma

In a BIBD, each variety is contained in

$$r = \frac{\lambda(v-1)}{k-1}$$

blocks.



Block designs V

- Let x_i be any one of the varieties, and suppose that x_i is contained in r blocks

$$B_{i_1}, B_{i_2}, \ldots, B_{i_r}. \tag{10.1}$$

  Since each block contains k elements, each of these blocks contains $k-1$ varieties other than x_i.
- We now consider each of the $v-1$ pairs $\{x_i, y\}$ where y is a variety different from x_i, and for each such pair count the number of blocks in which both varieties are contained. Each pair $\{x_i, y\}$ is contained in λ blocks (these blocks must be λ of the blocks in (10.1)). Adding we get

$$\lambda(v-1).$$



Block designs VI

- On the other hand, each of the block in (10.1) contains $k-1$ pairs, one elements of which is x_i. Adding we now get

$$r(k-1).$$

Equating these two counts we get

$$\lambda(v-1) = r(k-1).$$

Hence x_i is contained in $\lambda(v-1)/(k-1)$ blocks. Thus each variety is contained in $\lambda(v-1)/(k-1)$ blocks. □

Corollary

In a BIBD, we have

$$bk = vr.$$



Block designs VII

Corollary

In a BIBD, we have

$$\lambda < r.$$

- We call
  - b: the number of blocks;
  - v: the number of varieties;
  - k: the number of varieties in each block;
  - r: the number of blocks containing each variety;
  - λ: the number of blocks containing each pair of varieties



Block designs VIII

the *parameters* of the BIBD.

Example

Is there a BIBD with parameters $b = 12$, $k = 4$, $v = 16$, and $r = 3$ (the parameter λ is not specified)?

Solution

If there is such a design its index

$$\lambda = \frac{r(k-1)}{v-1} = \frac{3 \times 3}{15} = \frac{9}{15}.$$

Since this is not an integer there can be no such design with four of its parameters as given.



Block designs IX

Example

A design with parameters $b = 12$, $v = 9$, $k = 3$, $r = 4$, and $\lambda = 1$. It is straightforward to check that this matrix

$$A = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 \end{bmatrix}$$

defines a BIBD with parameters given.



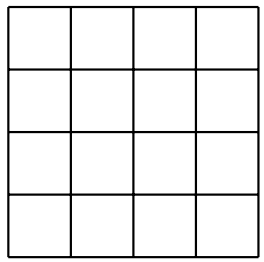
Block designs X

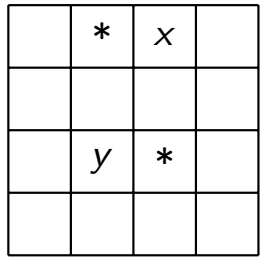
Example

Consider the squares of a 4-by-4 board as shown. Let the varieties be the 16 squares of the board. We define blocks as follows. For each square, we take the 6 squares which are either in its row or in its column (but not the square itself).

We thus have $b = 16$, $v = 16$, and $k = 6$. We also have $r = 6$. Let's take a pair of square x and y. There are three possibilities,

1. x and y are in the same row. Then x and y are together in the 2 blocks determined by the other 2 squares in their row.
2. x and y are in the same column. Then x and y are together in the 2 blocks determined by the other 2 squares in their column.
3. x and y are in different rows and in different columns. Then x and y are together in 2 blocks, one determined by the square at the intersection of the row of x and the column of y, the other by the intersection of the column of x and the row of y.







Block designs XI

Each pair of varieties is together in 2 blocks we have a BIBD with $\lambda = 2$.

Theorem

Fisher's inequality – In a BIBD, $b \geq v$.

- A BIBD for which for which the number b of blocks equals the number v of varieties, is called *symmetric*, and this is shorten to SBIBD. Thus, the parameters associated with an SBIBD are as follows,
  - v: the number of blocks;
  - v: the number of varieties;
  - k: the number of varieties in each block;
  - k: the number of blocks containing each variety;
  - λ: the number of blocks containing each pair of varieties, where

$$\lambda = \frac{k(k-1)}{v-1}. \tag{10.2}$$



Block designs XII

- Let $v \geq 2$ be an integer, and consider the set of integers mod v,

$$Z_v = \{0, 1, 2, \ldots, v-1\}$$

whose addition and multiplication are denoted by the usual symbols $+$ and $\times$. Let $B = \{i_1, i_2, \ldots, i_k\}$ be a subset of Z_v consisting of k integers. For each integer j in Z_v, we define

$$B + j = \{i_1 + j, i_2 + j, \ldots, i_k + j\}$$

to be the subset of Z_v obtained by adding mod v the integer j to each of the integers in B. The set $B + j$ also contain k integers. The v sets

$$B = B + 0, B + 1, \ldots, B + v - 1$$

so obtained are called the *blocks developed from the block* B and B is called the *starter block*.



Block designs XIII

Example

Let $v = 7$ and consider

$$Z_7 = \{0, 1, 2, 3, 4, 5, 6\}.$$

Now consider the starter block $B = \{0, 1, 3\}$. Then we have

$$B + 0 = \{0, 1, 3\},$$
$$B + 1 = \{1, 2, 4\},$$
$$B + 2 = \{2, 3, 5\},$$
$$B + 3 = \{3, 4, 6\},$$
$$B + 4 = \{4, 5, 0\},$$
$$B + 5 = \{5, 6, 1\},$$
$$B + 6 = \{6, 0, 2\}.$$



Block designs XIV

This is a BIBD, an SBIBD too, indeed, the same one in the introductory example of this section. $b = v = 7$, $k = r = 2$, and $\lambda = 1$.

Example

Let $v = 7$ as in the example above, but now let the starter block be $B = \{0, 1, 4\}$. Then we have

$$B + 0 = \{0, 1, 4\},$$
$$B + 1 = \{1, 2, 5\},$$
$$B + 2 = \{2, 3, 6\},$$
$$B + 3 = \{3, 4, 0\},$$
$$B + 4 = \{4, 5, 1\},$$
$$B + 5 = \{5, 6, 2\},$$
$$B + 6 = \{6, 0, 3\}.$$

We do not obtain a BIBD in this case.



Block designs XV

- Let B be a subset of k integers in Z_v. Then B is called a *difference set mod* v, provided that each nonzero integer in Z_v occurs the same number λ of times among the $k(k-1)$ differences among distinct elements of B (in both orders): $x - y (x, y \in B; x \neq y)$.

Example

Let $v = 7$ and $k = 3$. Consider $B = \{0, 1, 3\}$ and $B = \{0, 1, 4\}$, respectively. The subtraction tables for the integers in B are the following

−	0	1	3
0	0	6	4
1	1	0	5
3	3	2	0

−	0	1	4
0	0	6	3
1	1	0	4
4	4	3	0

One is a difference set, the other is not.



Block designs XVI

Theorem

Let B be a subset of $k < v$ elements of Z_v that forms a difference set mod v. Then the blocks developed from B as a starter block form an SBIBD with index $\lambda = \frac{k(k-1)}{v-1}$.



Block designs XVII

Example

Find a difference set of size 5 in Z_{11}, and use it as a starter block in order to construct an SBIBD.



Block designs XVII

Example

Find a difference set of size 5 in Z_{11}, and use it as a starter block in order to construct an SBIBD. Let $B = \{0, 2, 3, 4, 8\}$.



Block designs XVII

Example

Find a difference set of size 5 in Z_{11}, and use it as a starter block in order to construct an SBIBD. Let $B = \{0, 2, 3, 4, 8\}$.

$-$	0	2	3	4	8
0	0	9	8	7	3
2	2	0	10	9	5
3	3	1	0	10	6
4	4	2	1	0	7
8	8	6	5	4	0



Block designs XVII

Example

Find a difference set of size 5 in Z_{11}, and use it as a starter block in order to construct an SBIBD. Let $B = \{0, 2, 3, 4, 8\}$.

$-$	0	2	3	4	8
0	0	9	8	7	3
2	2	0	10	9	5
3	3	1	0	10	6
4	4	2	1	0	7
8	8	6	5	4	0

Each nonzero integer in Z_{11} occurs twice as a difference and B is a difference set.



Block designs XVII

Example

Find a difference set of size 5 in Z_{11}, and use it as a starter block in order to construct an SBIBD. Let $B = \{0, 2, 3, 4, 8\}$.

$-$	0	2	3	4	8
0	0	9	8	7	3
2	2	0	10	9	5
3	3	1	0	10	6
4	4	2	1	0	7
8	8	6	5	4	0

$$
\begin{aligned}
B + 0 &= \{0, 2, 3, 4, 8\} \\
B + 1 &= \{1, 3, 4, 5, 9\} \\
B + 2 &= \{2, 4, 5, 6, 10\} \\
B + 3 &= \{0, 3, 5, 6, 7\} \\
B + 4 &= \{1, 4, 6, 7, 8\} \\
B + 5 &= \{2, 5, 7, 8, 9\} \\
B + 6 &= \{3, 6, 8, 9, 10\} \\
B + 7 &= \{0, 4, 7, 9, 10\} \\
B + 8 &= \{0, 1, 5, 8, 10\} \\
B + 9 &= \{0, 1, 2, 6, 9\} \\
B + 10 &= \{1, 2, 3, 7, 10\}.
\end{aligned}
$$

Each nonzero integer in Z_{11} occurs twice as a difference and B is a difference set.



Block designs XVII

Example

Find a difference set of size 5 in Z_{11}, and use it as a starter block in order to construct an SBIBD. Let $B = \{0, 2, 3, 4, 8\}$.

$-$	0	2	3	4	8
0	0	9	8	7	3
2	2	0	10	9	5
3	3	1	0	10	6
4	4	2	1	0	7
8	8	6	5	4	0

$$
\begin{aligned}
B + 0 &= \{0, 2, 3, 4, 8\} \\
B + 1 &= \{1, 3, 4, 5, 9\} \\
B + 2 &= \{2, 4, 5, 6, 10\} \\
B + 3 &= \{0, 3, 5, 6, 7\} \\
B + 4 &= \{1, 4, 6, 7, 8\} \\
B + 5 &= \{2, 5, 7, 8, 9\} \\
B + 6 &= \{3, 6, 8, 9, 10\} \\
B + 7 &= \{0, 4, 7, 9, 10\} \\
B + 8 &= \{0, 1, 5, 8, 10\} \\
B + 9 &= \{0, 1, 2, 6, 9\} \\
B + 10 &= \{1, 2, 3, 7, 10\}.
\end{aligned}
$$

Each nonzero integer in Z_{11} occurs twice as a difference and B is a difference set. We obtain an SBIBD with $b = v = 11$, $k = r = 5$, and $\lambda = 2$.



Steiner triple systems I

Example

A BIBD with $v = 6$, $k = 2$, and $\lambda = 1$ is given by

$$\begin{array}{ccccc} \{0,1\} & \{0,2\} & \{0,3\} & \{0,4\} & \{0,5\} \\ \{1,2\} & \{1,3\} & \{1,4\} & \{1,5\} & \{2,3\} \\ \{2,4\} & \{2,5\} & \{3,4\} & \{3,5\} & \{4,5\} \end{array}$$

To get a BIBD with $\lambda = 2$, simply take each of the blocks twice. To get one with $\lambda = 3$, take each of the blocks three times.

- BIBD's with block size 2 are trivial. The smallest (in terms of block size) interesting case occurs when $k = 3$.
- Balanced block designs with block size $k = 3$ are called *Steiner triple systems*.
- The first example given in the previous section is a Steiner triple system. This is the only instance of a Steiner triple system which forms an SBIBD.



Steiner triple systems II

- Another example of a Steiner triple system is obtained by taking $v = 3$ varieties 0, 1, and 2 and one block $\{0, 1, 2\}$. We thus have $b = 1$ and $\lambda = 1$. This Steiner triple system is not an incomplete design since $v = k = 3$.

Example

A Steiner triple system of index $\lambda = 1$ with 9 varieties is given by

$$\begin{array}{lll} \{0,1,2\} & \{3,4,5\} & \{6,7,8\} \\ \{0,3,6\} & \{1,4,7\} & \{2,5,8\} \\ \{0,4,8\} & \{2,3,7\} & \{1,5,6\} \\ \{0,5,7\} & \{1,3,8\} & \{2,4,6\} \end{array}$$



Steiner triple systems III

Theorem

Let B be a Steiner triple system with parameters b, v, $k = 3$, r, λ. Then

$$r = \frac{\lambda(v-1)}{2} \tag{10.3}$$

and

$$b = \frac{\lambda v(v-1)}{6}. \tag{10.4}$$

If the index is $\lambda = 1$, then there is a nonnegative integer n such that $v = 6n + 1$ or $v = 6n + 3$.

Theorem

If there are Steiner triple systems of index $\lambda = 1$ with v and w varieties, respectively, then there is a Steiner triple system of index $\lambda = 1$ with vw varieties.



Steiner triple systems IV

Let $\mathcal{B}_1$ be a Steiner triple system of index $\lambda = 1$ with the v varieties $a_1, a_2, \ldots, a_v$ and let $\mathcal{B}_2$ be a Steiner triple system of index $\lambda = 1$ with the w varieties $b_1, b_2, \ldots, b_w$.

$$\begin{array}{c} \\ a_1 \\ a_2 \\ \vdots \\ a_v \end{array} \begin{array}{c} \begin{array}{cccc} b_1 & b_2 & \cdots & b_w \end{array} \\ \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1w} \\ c_{21} & c_{22} & \cdots & c_{2w} \\ \vdots & \vdots & \ddots & \vdots \\ c_{v1} & c_{v2} & \cdots & c_{vw} \end{bmatrix} \end{array} \tag{10.5}$$

Define a set $\mathcal{B}$ of triples of the elements of X. Let $\{c_{\alpha\mu}, c_{\beta\nu}, c_{\gamma\sigma}\}$ is a triple of $\mathcal{B}$ is and only if one of the following holds.

1. $\mu = \nu = \sigma$, and $\{a_\alpha, a_\beta, a_\gamma\}$ is a triple in $\mathcal{B}_1$.
2. $\alpha = \beta = \gamma$, and $\{b_\mu, b_\nu, b_\sigma\}$ is a triple in $\mathcal{B}_2$.
3. μ, ν, and σ are all different and $\{a_\alpha, a_\beta, a_\gamma\}$ is a triple in $\mathcal{B}_1$, and α, β, and γ are all different and $\{b_\mu, b_\nu, b_\sigma\}$ is a triple in $\mathcal{B}_2$. □



Steiner triple systems V

Example

The simplest instance is that obtained by choosing $\mathcal{B}_1$ and $\mathcal{B}_2$ to be Steiner triple system with 3 varieties. The result should be a Steiner triple system with $3 \times 3 = 9$ varieties.

- Let $\mathcal{B}_1$ be the Steiner triple system with the 3 varieties a_1, a_2, a_3 and unique triple $\{a_1, a_2, a_3\}$, and let $\mathcal{B}_2$ be the Steiner triple system with the 3 varieties b_1, b_2, b_3 and unique triple $\{b_1, b_2, b_3\}$.
- We consider the set X of 9 varieties comprising the entries of the following array,

$$\begin{array}{c} \\ a_1 \\ a_2 \\ a_3 \end{array} \begin{array}{c} \begin{array}{ccc} b_1 & b_2 & b_3 \end{array} \\ \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix} \end{array}$$

We have the following set of 12 triples comprising a Steiner triple system of index 1 with 9 varieties.



Steiner triple systems VI

1. The entries in each of the 3 rows,

$$\{c_{11}, c_{12}, c_{13}\}, \quad \{c_{21}, c_{22}, c_{23}\}, \quad \{c_{31}, c_{32}, c_{33}\}.$$

2. The entries in each of the 3 columns,

$$\{c_{11}, c_{21}, c_{31}\}, \quad \{c_{12}, c_{22}, c_{32}\}, \quad \{c_{13}, c_{23}, c_{33}\}.$$

3. Three entries, no two from the same row or column,

$$\{c_{11}, c_{22}, c_{33}\}, \quad \{c_{12}, c_{23}, c_{31}\}, \quad \{c_{13}, c_{21}, c_{32}\}$$
$$\{c_{13}, c_{22}, c_{31}\}, \quad \{c_{12}, c_{21}, c_{33}\}, \quad \{c_{11}, c_{23}, c_{32}\}.$$

If we replace $c_{11}, c_{21}, c_{31}, c_{12}, c_{22}, c_{32}, c_{13}, c_{23}, c_{33}$ by 0, 1, 2, 3, 4, 5, 6, 7, 8, respectively, we obtain the Steiner triple system $\mathcal{B}$ with 9 varieties given earlier in this section.

$$\begin{array}{cccc} \{0,1,2\} & \{0,3,6\} & \{0,4,8\} & \{0,5,7\} \\ \{3,4,5\} & \{1,4,7\} & \{2,3,7\} & \{1,3,8\} \\ \{6,7,8\} & \{2,5,8\} & \{1,5,6\} & \{2,4,6\} \end{array} \tag{10.6}$$



Steiner triple systems VII

- The columns of (10.6) partition the triples of $\mathcal{B}$ into parts so that each variety occurs in exactly one triple in each part. A Steiner triple system of index $\lambda = 1$ with this property is called *resolvable* and each part is called a *resolvable class*.

Kirkman's schoolgirl problem

A schoolmistress takes her class of 15 girls on a daily walk. The girls are arranged in 5 rows, with 3 girls in each row, so that each girl has 2 companions. Is it possible to plan a walk for 7 consecutive days so that no girl will walk with any of her classmates in a triplet more than once?

- A resolvable Steiner triple system of index $\lambda = 1$ is also called a *Kirkman's triple system*. Ray-Chaudhuri et al. (1971) showed how to construct such a system for all n.

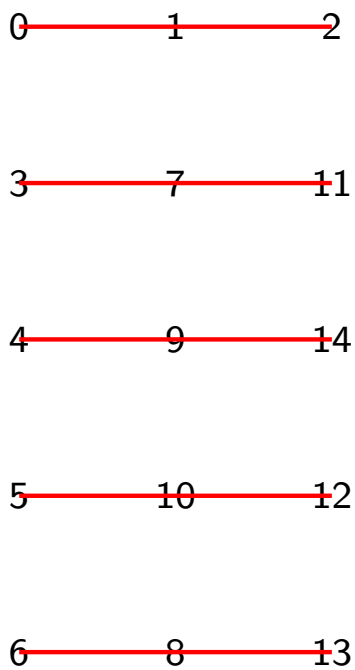


Steiner triple systems VIII

Example

Solution of Kirkman's schoolgirl problem

$\{0,1,2\}$ $\{0,3,4\}$ $\{0,5,6\}$ $\{0,7,8\}$
$\{3,7,11\}$ $\{1,7,9\}$ $\{1,8,10\}$ $\{1,11,13\}$
$\{4,9,14\}$ $\{2,12,13\}$ $\{2,11,14\}$ $\{2,4,5\}$
$\{5,10,12\}$ $\{5,8,14\}$ $\{3,9,13\}$ $\{3,10,14\}$
$\{6,8,13\}$ $\{6,10,11\}$ $\{4,7,12\}$ $\{6,9,12\}$
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$\{2,3,6\}$ $\{2,8,9\}$ $\{2,7,10\}$
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$\{5,7,13\}$ $\{6,7,14\}$ $\{5,9,11\}$



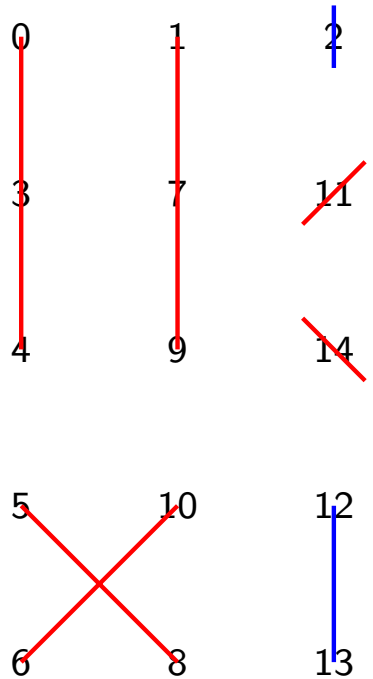
Steiner triple systems VIII

Example

Solution of Kirkman's schoolgirl problem

$\{0,1,2\}$ $\{0,3,4\}$ $\{0,5,6\}$ $\{0,7,8\}$
$\{3,7,11\}$ $\{1,7,9\}$ $\{1,8,10\}$ $\{1,11,13\}$
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$\{0,9,10\}$ $\{0,11,12\}$ $\{0,13,14\}$
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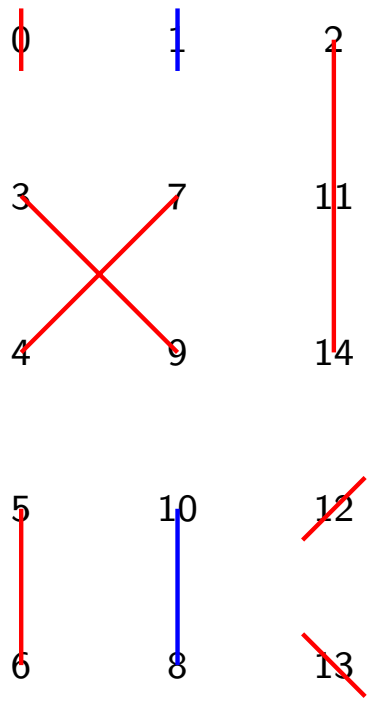


Steiner triple systems VIII

Example

Solution of Kirkman's schoolgirl problem

$\{0,1,2\}$	$\{0,3,4\}$	$\{0,5,6\}$	$\{0,7,8\}$
$\{3,7,11\}$	$\{1,7,9\}$	$\{1,8,10\}$	$\{1,11,13\}$
$\{4,9,14\}$	$\{2,12,13\}$	$\{2,11,14\}$	$\{2,4,5\}$
$\{5,10,12\}$	$\{5,8,14\}$	$\{3,9,13\}$	$\{3,10,14\}$
$\{6,8,13\}$	$\{6,10,11\}$	$\{4,7,12\}$	$\{6,9,12\}$
	$\{0,9,10\}$	$\{0,11,12\}$	$\{0,13,14\}$
	$\{1,12,14\}$	$\{1,3,5\}$	$\{1,4,6\}$
	$\{2,3,6\}$	$\{2,8,9\}$	$\{2,7,10\}$
	$\{4,8,11\}$	$\{4,10,13\}$	$\{3,8,12\}$
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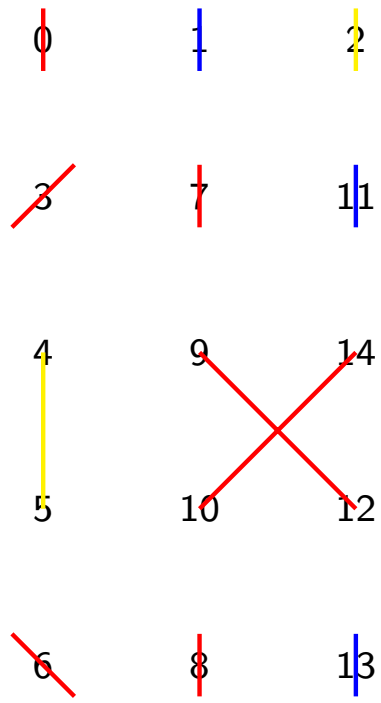


Steiner triple systems VIII

Example

Solution of Kirkman's schoolgirl problem

$\{0,1,2\}$	$\{0,3,4\}$	$\{0,5,6\}$	$\{0,7,8\}$
$\{3,7,11\}$	$\{1,7,9\}$	$\{1,8,10\}$	$\{1,11,13\}$
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$\{5,10,12\}$	$\{5,8,14\}$	$\{3,9,13\}$	$\{3,10,14\}$
$\{6,8,13\}$	$\{6,10,11\}$	$\{4,7,12\}$	$\{6,9,12\}$
	$\{0,9,10\}$	$\{0,11,12\}$	$\{0,13,14\}$
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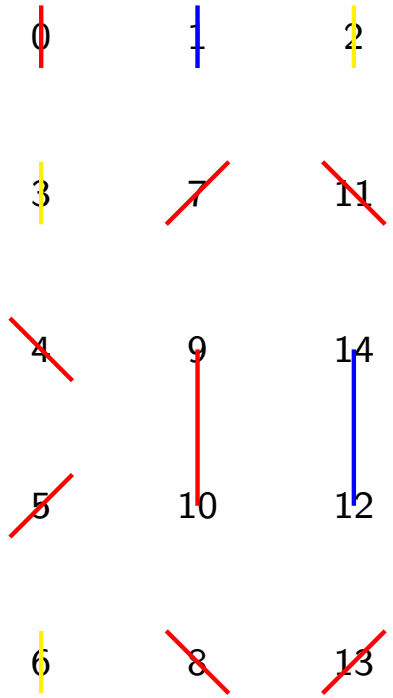
Steiner triple systems VIII

Example

Solution of Kirkman's schoolgirl problem

$\{0,1,2\}$ $\{0,3,4\}$ $\{0,5,6\}$ $\{0,7,8\}$
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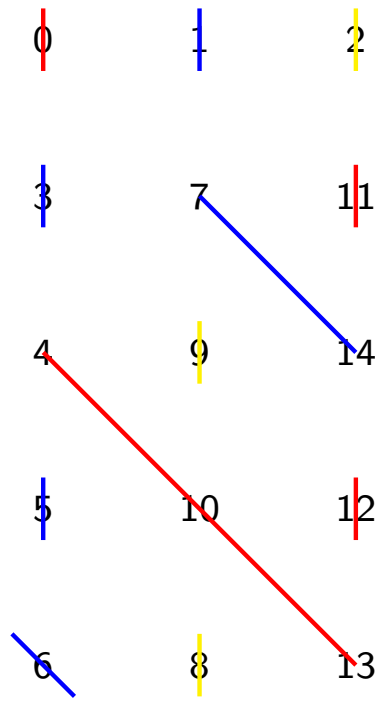
Steiner triple systems VIII

Example

Solution of Kirkman's schoolgirl problem

$\{0,1,2\}$ $\{0,3,4\}$ $\{0,5,6\}$ $\{0,7,8\}$
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$\{6,8,13\}$ $\{6,10,11\}$ $\{4,7,12\}$ $\{6,9,12\}$

$\{0,9,10\}$ $\{0,11,12\}$ $\{0,13,14\}$
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$\{5,7,13\}$ $\{6,7,14\}$ $\{5,9,11\}$



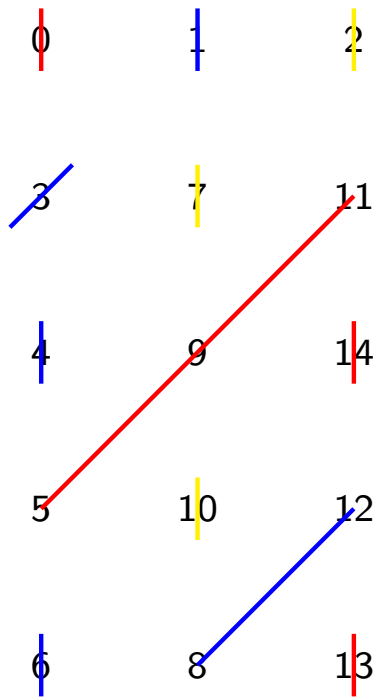
Steiner triple systems VIII

Example

Solution of Kirkman's schoolgirl problem

$\{0,1,2\}$ $\{0,3,4\}$ $\{0,5,6\}$ $\{0,7,8\}$
$\{3,7,11\}$ $\{1,7,9\}$ $\{1,8,10\}$ $\{1,11,13\}$
$\{4,9,14\}$ $\{2,12,13\}$ $\{2,11,14\}$ $\{2,4,5\}$
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$\{4,8,11\}$ $\{4,10,13\}$ $\{3,8,12\}$
$\{5,7,13\}$ $\{6,7,14\}$ $\{5,9,11\}$



Latin squares I

- Let n be a positive integer and let S be a set of n distinct elements. A *Latin square of order n*, based on the set S, is an n-by-n array, each of whose entries is an element of S such that each of the n elements of S occurs once (and hence exactly once) in each row and once in each column.
- Usually we take S to be $Z_n = \{0, 1, \ldots, n-1\}$. In this case, we number the rows and columns of the Latin square as $0, 1, \ldots, n-1$, rather than the more conventional $1, 2, \ldots, n$.

Example

Latin squares with $n = 1$, 2, 3, and 4 are as the following,

$$[0], \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 0 \\ 2 & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 & 2 & 3 \\ 1 & 2 & 3 & 0 \\ 2 & 3 & 0 & 1 \\ 3 & 0 & 1 & 2 \end{bmatrix}.$$



Latin squares II

- Consider a Latin square of order n based on Z_n, and let k be any element of Z_n. Then k occurs n times in A, once in each row and once in each column. Let $A(k)$ be the set of positions occupied by k's $(k = 0, 1, \ldots, n-1)$. Then $A(0), A(1), \ldots, A(n-1)$ is a partition of the set of n^2 positions of the board.
- We can always bring a Latin square to *standard form*, whereby in row 0 the integers $0, 1, \ldots, n-1$ occur in their natural order.

Theorem

Let n be a positive integer. Let A be the n-by-n array whose entry a_{ij} in row i and column j is

$$a_{ij} = i + j \text{ (addition mod } n), \quad (i, j = 0, 1, \ldots, n-1).$$

Then A is a Latin square of order n based on Z_n.



Latin squares III

example

We consider Z_5, the integers mode 5. 3 has a multiplication inverse in Z_5, in fact $3 \times 2 = 1$ in Z_5. We construct a 5-by-5 array whose entry in row i and column j is $a_{ij} = 3 \times i + j$. The result is

$$\begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{ccccc} 0 & 1 & 2 & 3 & 4 \end{array} \\ \left[\begin{array}{ccccc} 0 & 1 & 2 & 3 & 4 \\ 3 & 4 & 0 & 1 & 2 \\ 1 & 2 & 3 & 4 & 0 \\ 4 & 0 & 1 & 2 & 3 \\ 2 & 3 & 4 & 0 & 1 \end{array}\right] \end{array}$$

Inspection reveals that we have a Latin square of order 5.



Latin squares IV

Theorem

Let n be a positive integer and let r be a nonzero integer in Z_n such that the GCD of r and n is 1. Let A be the n-by-n array whose entry a_{ij} in row i and column j is

$$a_{ij} = r \times i + j \text{ (arithmetic mod } n), \quad (i, j = 0, 1, \ldots, n-1).$$

Then A is a Latin square of order n based on Z_n.

- The Latin square of order n constructed by using an integer r with a multiplicative inverse in Z_n, will be denoted by

  $$L_n^r.$$

  If r does not have a multiplicative inverse, then the resulting array L_n^r will not be a Latin square.



Latin squares V

- Let

$$R_n = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ 1 & 1 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ n-1 & n-1 & \cdots & n-1 \end{bmatrix} \tag{10.7}$$

and

$$S_n = \begin{bmatrix} 0 & 1 & \cdots & n-1 \\ 0 & 1 & \cdots & n-1 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 1 & \cdots & n-1 \end{bmatrix} \tag{10.8}$$

by two n-by-n arrays based on Z_n with identical columns and rows, respectively, as shown. Let A be any n-by-n array based on Z_n. Then A is a Latin square if and only the following condition is satisfied,



Latin squares VI

1. When the arrays R_n and A are juxtaposed to form an array $R_n \times A$, the set of ordered pairs thus obtained equals the set of *all* ordered pairs (i, j) that can be formed using the elements of Z_n;
2. When the arrays S_n and A are juxtaposed to form an array $S_n \times A$, the set of ordered pairs thus obtained equals the set of *all* ordered pairs (i, j) that can be formed using the elements of Z_n;

Alternatively, we say that the ordered pairs in $R_n \times A$ are all distinct, and the ordered pair in $S_n \times A$ are all distinct.



Latin squares VII

Example

A Latin square of order 3,

$$\begin{bmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ 2 & 2 & 2 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 0 \\ 2 & 0 & 1 \end{bmatrix} \to \begin{bmatrix} (0,0) & (0,1) & (0,2) \\ (1,1) & (1,2) & (1,0) \\ (2,2) & (2,0) & (2,1) \end{bmatrix},$$

$$\begin{bmatrix} 0 & 1 & 2 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 0 \\ 2 & 0 & 1 \end{bmatrix} \to \begin{bmatrix} (0,0) & (1,1) & (2,2) \\ (0,1) & (1,2) & (2,0) \\ (0,2) & (1,0) & (2,1) \end{bmatrix}.$$

In each of the two juxtaposed arrays, each ordered pair occurs exactly once.



Latin squares VIII

- Let A and B be Latin squares based, for instance, on the integers in Z_n. Then A and B are called *orthogonal*, provided that in the juxtaposed array $A \times B$, each of the ordered pairs (i, j) of integers in Z_n occurs exactly once.

Example

The following two Latin squares of order 4 are orthogonal as is seen by examining their juxtaposed array,

$$\begin{bmatrix} 0 & 1 & 2 & 3 \\ 1 & 0 & 3 & 2 \\ 2 & 3 & 0 & 1 \\ 3 & 2 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 2 & 3 \\ 3 & 2 & 1 & 0 \\ 1 & 0 & 3 & 2 \\ 2 & 3 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} (0,0) & (1,1) & (2,2) & (3,3) \\ (1,3) & (0,2) & (3,1) & (2,0) \\ (2,1) & (3,0) & (0,3) & (1,2) \\ (3,2) & (2,3) & (1,0) & (0,1) \end{bmatrix}.$$

- Orthogonal Latin squares have application to the design of experiments in which variational differences need to be kept at a minimum in order to be able to draw meaningful conclusion.



Latin squares IX

Example

It is desired to test the effects of various quantities of water and various types (or qualities) of fertilizer on the yield of wheat on a certain type of soil. Suppose there are n quantities of water and n types of fertilizer to be tested, so that there are n^2 possible combinations of water and fertilizer.
We have at our disposal a rectangular field which is subdivided into n^2 plots, one for each of the n^2 possible water-fertilizer combinations. There is no reason to expect the soil fertilizer is the same throughout the field. Thus it may very well be that the first row is of high fertility, and therefore a higher yield of wheat will occur which is not solely due to the quantity of water and the type of fertilizer used on it. We are likely to minimize the influence of soil fertility on the yield of wheat if we insist that each quantity of water occur no more than once in any row and in any column, and similarly that each type of fertilizer occur no more than once in any row and in any column. Thus the application of the n quantities of water on the n^2 plots should determine a Latin square A of order n, and also the application of the n types of fertilizer



Latin squares X

should determine a Latin square B of order n. Since all n^2 possible water-fertilizer combinations are to be treated, when the two Latin squares A and B are juxtaposed all n^2 combinations should occur once. Thus the Latin squares A and B are to be orthogonal. Thus two orthogonal Latin squares of order n, one for the application of the n quantities of water and one for the n types of fertilizer, determine a design for an experiment to test the effects of water and fertilizer on the production of wheat.

- Let $A_1, A_2, \ldots, A_k$ be Latin squares base on Z_n, we say that $A_1, A_2, \ldots, A_k$ are *mutually orthogonal*, provided that each pair A_i, A_j $(i \neq j)$ of them is orthogonal. We refer to mutually orthogonal Latin squares as *MOLS*.

Theorem

Let n be a prime number. Then $L_n^1, L_n^2, \ldots, L_n^{n-1}$ are $n-1$ MOLS of order n.



Latin squares XI

- Let F be a finite field with $n = p^k$ elements for some prime p and positive integer k. Let

$$\alpha_0 = 0, \alpha_1, \ldots, \alpha_{n-1}$$

be the elements of F with α_0, as indicated, the zero element of F. Consider any nonzero elements α_r, $(r \neq 0)$ of F, and define an n-by-n array A such that the element a_{ij} in row i and column j of A is

$$a_{ij} = \alpha_r \times \alpha_i + \alpha_j, \quad (i, j = 0, 1, \ldots, n-1),$$

where the arithmetic is that of the field F.

Theorem

Let $n = p^k$ be a integer that is a power of a prime number p. Then there exist $n-1$ MOLS of order n. In fact, the $n-1$ Latin squares $L_n^{\alpha_1}, L_n^{\alpha_2}, \ldots, L_n^{\alpha_{n-1}}$ of order n constructed from a finite field with $n = p^k$ elements are $n-1$ MOLS of order n.

Latin squares XII

Example

Three MOLS of order 4

$$L_4^1 = \begin{bmatrix} 0 & 1 & i & 1+i \\ 1 & 0 & 1+i & i \\ i & 1+i & 0 & 1 \\ 1+i & i & 1 & 0 \end{bmatrix}, \quad L_4^i = \begin{bmatrix} 0 & 1 & i & 1+i \\ i & 1+i & 0 & 1 \\ 1+i & i & 1 & 0 \\ 1 & 0 & 1+i & i \end{bmatrix},$$

$$L_4^{1+i} = \begin{bmatrix} 0 & 1 & i & 1+i \\ 1+i & i & 1 & 0 \\ 1 & 0 & 1+i & i \\ i & 1+i & 0 & 1 \end{bmatrix}.$$

Theorem

Let $n \geq 2$ be an integer, and let $A_1, A_2, \ldots, A_k$ be k MOLS of order n. Then $k \leq n-1$; that is, the largest number of MOLS of order n is at most $n-1$.



Latin squares XIII

- For n a positive integer, Let $N(n)$ be the largest number of MOLS of order n.

Theorem

$N(n) \geq 2$ for each odd integer n.

Theorem

If there is a pair of MOLS of order m and there is a pair of MOLS of order k, then there is a pair of MOLS of order mk. More general,

$$N(mk) \geq \min\{N(m), N(k)\}.$$



Latin squares XIV

Example

Consider the two MOLS of order 3 and the two MOLS of order 4 given by

$$A_1 = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 0 \\ 2 & 0 & 1 \end{bmatrix}, A_2 = \begin{bmatrix} 0 & 2 & 1 \\ 1 & 0 & 2 \\ 2 & 1 & 0 \end{bmatrix}, B_1 = \begin{bmatrix} 0 & 1 & 2 & 3 \\ 1 & 0 & 3 & 2 \\ 2 & 3 & 0 & 1 \\ 3 & 2 & 1 & 0 \end{bmatrix}, B_2 = \begin{bmatrix} 0 & 1 & 2 & 3 \\ 2 & 3 & 0 & 1 \\ 3 & 2 & 1 & 0 \\ 1 & 0 & 3 & 2 \end{bmatrix}$$



Latin squares XV

We now form the 12-by-12 arrays $A_1 \otimes B_1$ and $A_2 \otimes B_2$

$$\begin{bmatrix}
(0,0) & (0,1) & (0,2) & (0,3) & (1,0) & (1,1) & (1,2) & (1,3) & (2,0) & (2,1) & (2,2) & (2,3) \\
(0,1) & (0,0) & (0,3) & (0,2) & (1,1) & (1,0) & (1,3) & (1,2) & (2,1) & (2,0) & (2,3) & (2,2) \\
(0,2) & (0,3) & (0,0) & (0,1) & (1,2) & (1,3) & (1,0) & (1,1) & (2,2) & (2,3) & (2,0) & (2,1) \\
(0,3) & (0,2) & (0,1) & (0,0) & (1,3) & (1,2) & (1,1) & (1,0) & (2,3) & (2,2) & (2,1) & (2,0) \\
(1,0) & (1,1) & (1,2) & (1,3) & (2,0) & (2,1) & (2,2) & (2,3) & (0,0) & (0,1) & (0,2) & (0,3) \\
(1,1) & (1,0) & (1,3) & (1,2) & (2,1) & (2,0) & (2,3) & (2,2) & (0,1) & (0,0) & (0,3) & (0,2) \\
(1,2) & (1,3) & (1,0) & (1,1) & (2,2) & (2,3) & (2,0) & (2,1) & (0,2) & (0,3) & (0,0) & (0,1) \\
(1,3) & (1,2) & (1,1) & (1,0) & (2,3) & (2,2) & (2,1) & (2,0) & (0,3) & (0,2) & (0,1) & (0,0) \\
(2,0) & (2,1) & (2,2) & (2,3) & (0,0) & (0,1) & (0,2) & (0,3) & (1,0) & (1,1) & (1,2) & (1,3) \\
(2,1) & (2,0) & (2,3) & (2,2) & (0,1) & (0,0) & (0,3) & (0,2) & (1,1) & (1,0) & (1,3) & (1,2) \\
(2,2) & (2,3) & (2,0) & (2,1) & (0,2) & (0,3) & (0,0) & (0,1) & (1,2) & (1,3) & (1,0) & (1,1) \\
(2,3) & (2,2) & (2,1) & (2,0) & (0,3) & (0,2) & (0,1) & (0,0) & (1,3) & (1,2) & (1,1) & (1,0)
\end{bmatrix}$$



Latin squares XV

We now form the 12-by-12 arrays $A_1 \otimes B_1$ and $A_2 \otimes B_2$

$$\begin{bmatrix}
0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 \\
1 & 0 & 3 & 2 & 5 & 4 & 7 & 6 & 9 & 8 & 11 & 10 \\
2 & 3 & 0 & 1 & 6 & 7 & 4 & 5 & 10 & 11 & 8 & 9 \\
3 & 2 & 1 & 0 & 7 & 6 & 5 & 4 & 11 & 10 & 9 & 8 \\
4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 0 & 1 & 2 & 3 \\
5 & 4 & 7 & 6 & 9 & 8 & 11 & 10 & 1 & 0 & 3 & 2 \\
6 & 7 & 4 & 5 & 10 & 11 & 8 & 9 & 2 & 3 & 0 & 1 \\
7 & 6 & 5 & 4 & 11 & 10 & 9 & 8 & 3 & 2 & 1 & 0 \\
8 & 9 & 10 & 11 & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\
9 & 8 & 11 & 10 & 1 & 0 & 3 & 2 & 5 & 4 & 7 & 6 \\
10 & 11 & 8 & 9 & 2 & 3 & 0 & 1 & 6 & 7 & 4 & 5 \\
11 & 10 & 9 & 8 & 3 & 2 & 1 & 0 & 7 & 6 & 5 & 4
\end{bmatrix}$$



Latin squares XV

We now form the 12-by-12 arrays $A_1 \otimes B_1$ and $A_2 \otimes B_2$

$$\begin{bmatrix}
0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 \\
1 & 0 & 3 & 2 & 5 & 4 & 7 & 6 & 9 & 8 & 11 & 10 \\
2 & 3 & 0 & 1 & 6 & 7 & 4 & 5 & 10 & 11 & 8 & 9 \\
3 & 2 & 1 & 0 & 7 & 6 & 5 & 4 & 11 & 10 & 9 & 8 \\
4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 0 & 1 & 2 & 3 \\
5 & 4 & 7 & 6 & 9 & 8 & 11 & 10 & 1 & 0 & 3 & 2 \\
6 & 7 & 4 & 5 & 10 & 11 & 8 & 9 & 2 & 3 & 0 & 1 \\
7 & 6 & 5 & 4 & 11 & 10 & 9 & 8 & 3 & 2 & 1 & 0 \\
8 & 9 & 10 & 11 & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\
9 & 8 & 11 & 10 & 1 & 0 & 3 & 2 & 5 & 4 & 7 & 6 \\
10 & 11 & 8 & 9 & 2 & 3 & 0 & 1 & 6 & 7 & 4 & 5 \\
11 & 10 & 9 & 8 & 3 & 2 & 1 & 0 & 7 & 6 & 5 & 4
\end{bmatrix},
\begin{bmatrix}
0 & 1 & 2 & 3 & 8 & 9 & 10 & 11 & 4 & 5 & 6 & 7 \\
2 & 3 & 0 & 1 & 10 & 11 & 8 & 9 & 6 & 7 & 4 & 5 \\
3 & 2 & 1 & 0 & 11 & 10 & 9 & 8 & 7 & 6 & 5 & 4 \\
1 & 0 & 3 & 2 & 9 & 8 & 11 & 10 & 5 & 4 & 7 & 6 \\
4 & 5 & 6 & 7 & 0 & 1 & 2 & 3 & 8 & 9 & 10 & 11 \\
6 & 7 & 4 & 5 & 2 & 3 & 0 & 1 & 10 & 11 & 8 & 9 \\
7 & 6 & 5 & 4 & 3 & 2 & 1 & 0 & 11 & 10 & 9 & 8 \\
5 & 4 & 7 & 6 & 1 & 0 & 3 & 2 & 9 & 8 & 11 & 10 \\
8 & 9 & 10 & 11 & 4 & 5 & 6 & 7 & 0 & 1 & 2 & 3 \\
10 & 11 & 8 & 9 & 6 & 7 & 4 & 5 & 2 & 3 & 0 & 1 \\
11 & 10 & 9 & 8 & 7 & 6 & 5 & 4 & 3 & 2 & 1 & 0 \\
9 & 8 & 11 & 10 & 5 & 4 & 7 & 6 & 1 & 0 & 3 & 2
\end{bmatrix}$$



Latin squares XVI

Theorem

Let $n \geq 2$ be an integer, and let

$$n = p_1^{e_1} \times p_2^{e_2} \times \cdots \times p_k^{e_k}$$

be the factorization of n into distinct prime numbers $p_1, p_2, \ldots, p_k$. Then

$$N(n) \geq \min\{p_i^{e_i} - 1 : i = 1, 2, \ldots, k\}.$$

Corollary

Let $n \geq 2$ be an integer that is not twice an odd number. Then there exists a pair of orthogonal Latin squares of order n.

The integers n for which the corollary does not guarantee the existence of a pair of MOLS of order n are the integers

$$2, 6, 10, 14, 18, \ldots, 4k + 2, \ldots. \tag{10.9}$$

It is easy to see that there is no pair of Latin squares of order 2.



Latin squares XVII

- Euler concluded, but did not prove, that there is no pair of orthogonal Latin squares of order 6, and he conjectured that no such pair existed for any of integers $6, 10, 14, 18, \ldots, 4k+2, \ldots$.
- By exhaustive enumeration Tary (1900/1901) proved that Euler conjecture was true for $n = 6$. Euler conjecture was false for all $n > 6$ Bose et al. (1960).
- Let $A_1, A_2, \ldots, A_{n-1}$ denote $n-1$ MOLS of order n. We use the $n+1$ arrays

$$R_n, S_n, A_1, A_2, \ldots, A_{n-1}, \tag{10.10}$$

where R_n and S_n are defined in (10.7) and (10.8), to construct a block design $\mathcal{B}$ with parameters

$$b = n^2 + n, v = n^2, k = n, r = n+1, \lambda = 1.$$

Recall that $A_i(k)$ denotes the set of positions of A_i occupied by k $(k = 0, 1, \ldots, n-1)$. A_i is a Latin square. $A_i(k)$ contains one position from each row and each column; in particular, no two positions in $A_i(k)$ belong to the same row or to the same column.

- We also use this notation for R_n and S_n. For instance, $R_n(0)$ denotes the set of positions of R_n that are occupied by 0's, and this set is the set of positions of row 0.



Latin squares XVIII

We take the set X of varieties to be the set of $v = n^2$ positions of an n-by-n array; that is,

$$X = \{(i,j), i,j = 0,1,\cdots,n-1\}.$$

Each of the $n+1$ arrays in (10.10) determines n blocks

$$R_n(0) \quad R_n(1) \quad \cdots \quad R_n(n-1) \tag{10.11}$$

$$S_n(0) \quad S_n(1) \quad \cdots \quad S_n(n-1) \tag{10.12}$$

$$\begin{array}{cccc} A_1(0) & A_1(1) & \cdots & A_1(n-1) \\ \vdots & \vdots & \cdots & \vdots \\ A_{n-1}(0) & A_{n-1}(1) & \cdots & A_{n-1}(n-1) \end{array} \tag{10.13}$$

Thus, we have $b = n \times (n+1) = n^2 + n$ blocks, each containing $k = n$ varieties. Let $\mathcal{B}$ denote this collection of blocks. $\mathcal{B}$ is hereby a BIBD. The design $\mathcal{B}$ constructed is *resolvable*. The collection of $n^2 + n$ blocks is partitioned into $n+1$ parts (*resolvability classes*) of n blocks each, and each resovability class is a partition of the n^2 varieties.

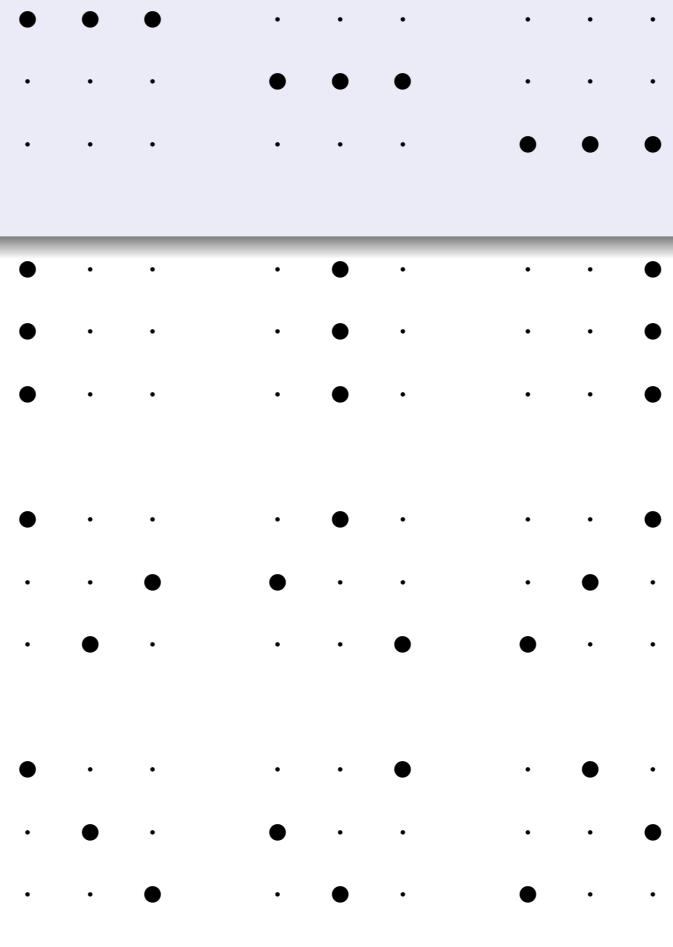


Latin squares XIX

Example

Construction of a BIBD using the two Latin squares of order 3

$$A_1 = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 0 \\ 2 & 0 & 1 \end{bmatrix}, A_2 = \begin{bmatrix} 0 & 2 & 1 \\ 1 & 0 & 2 \\ 2 & 1 & 0 \end{bmatrix}.$$



The varieties are the 9 positions of a 9-by-9 array, and the blocks are pictured geometrically by resolvability classes as follows,

- varities as *points*
- blocks as *lines*
- each of the resovability classes consists of 3 *parallel* lines
- each pair of varieties being together in exactly one block. Two points determine exactly one line

Latin squares XX

Theorem

Let $n \geq 2$ be an integer. If there exist $n-1$ MOLS of order n, then there exists a resolvable BIBD with parameters

$$b = n^2 + n, v = n^2, k = n, r = n + 1, \lambda = 1. \tag{10.14}$$

Conversely, if there exists a resolvable BIBD with parameters (10.14), then there exist $n-1$ MOLS of order n.

There are three natural ways to construct a Latin square of order n,

1. row-by-row,
2. column-by-column, and
3. element-by-element.

Will we ever get stuck?



Latin squares XXI

Let m and n be integers with $m \leq n$. An m-by-n *Latin rectangle*, based on the integers in Z_n, is an m-by-n array such that no integer is repeated in any row or in any column. If $m = n$, then our definition of a Latin rectangle is equivalent to that of a Latin square. We say that an m-by-n Latin rectangle L can be *completed*, provided it is possible to attach $n - m$ rows to L and obtain a Latin square L^* of order n. Such a Latin square L^* is called a *completion* of L.

Example

$$L = \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 3 & 4 & 0 & 1 & 2 \\ 4 & 0 & 3 & 2 & 1 \end{bmatrix}, \qquad L^* = \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 3 & 4 & 0 & 1 & 2 \\ 4 & 0 & 3 & 2 & 1 \\ 2 & 3 & 1 & 4 & 0 \\ 1 & 2 & 4 & 0 & 3 \end{bmatrix}$$



Latin squares XXII

Theorem

Let L be an m-by-n Latin rectangle base on Z_n with $m < n$. Then L has a completion.

- Consider an n-by-n array L in which some positions are unoccupied and other positions are occupied by one of the integers $\{0, 1, \cdots, n-1\}$. Suppose that, if an integer k occurs in L, then it occurs n times and no two k's belong to the same row or column. Then we call L a *semi-Latin square*. If m different integers occur in L, then we say L has *index* m.
- We say that a semi-Latin square L of order n can be *completed* to a Latin square, provided that it is possible to fill in the unoccupied positions in order to obtain a Latin square $L^{\#}$ of order n. Such a Latin square of $L^{\#}$ is called a *completion* of L.



Latin squares XXIII

Example

A semi-Latin square L of order 5 and index 3 and its completion $L^{\#}$

$$L = \begin{bmatrix} 1 & & 0 & & 2 \\ & 2 & 1 & & 0 \\ 0 & 1 & & 2 & \\ 2 & 0 & & 1 & \\ & & 2 & 0 & 1 \end{bmatrix}, \qquad L^{\#} = \begin{bmatrix} 1 & 4 & 0 & 3 & 2 \\ 3 & 2 & 1 & 4 & 0 \\ 0 & 1 & 4 & 2 & 3 \\ 2 & 0 & 3 & 1 & 4 \\ 4 & 3 & 2 & 0 & 1 \end{bmatrix}$$

Theorem

Let L be a semi-Latin square of order n and index m where $m < n$. Then L has a completion.



Latin squares XXIV

There is a one-to-one correspondence between m-by-n Latin rectangles and semi-Latin squares of order n and index m. Let L be an m-by$-n$ Latin rectangle and let the entry in position row i, column j be denoted by a_{ij}. We define an n-by-n array B by letting the entry b_{ij} in position row i, column j be k, provided that i occurs in column j of row k of L. Thus $b_{ij} = k$ is and only if $a_{kj} = i$.

Example

$$A = \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 0 & 2 \\ 1 & 0 & 4 & 2 & 3 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 2 & & 1 & \\ 2 & 0 & 1 & & \\ & & 0 & 2 & 1 \\ 1 & & & 0 & 2 \\ & 1 & 2 & & 0 \end{bmatrix}.$$

